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Report

The report will contain description of our work and included the features of the system and how each feature works, Algorithm we used and the manual to know how use it.

Manual

Frist: The initial menu the program will ask you to enter your choice you need to enter number 1 or 2.

```
===== Menu =====  
  
1. LOGIN  
2. QUIT  
-----  
Enter your choice (numeric) :_
```

In the case of choice number 2, that will exit the program.

In the case of choice number 1, that will require you to enter your username and password and if it true it will send you to the main menu.

```
===== Menu =====  
  
1. LOGIN  
2. QUIT  
-----  
Enter your choice (numeric) :1  
  
===== Login =====  
  
Enter username: aaa  
Enter password: 123a_
```

Second: The Main Menu will ask you to enter a number from 1 to 11.

```
===== Menu =====
```

1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT

```
-----  
Enter your choice (numeric) :_
```

In the case of choice number 1, that will send you to the **Add** account part that will require you to enter the new account number consist of 10 digits, the first and last name, Email address and the mobile number. At the end it will ask you if you want to save your changes in case of Yes just enter (y) and in case of you want to discard your changes enter (n).

```
===== Menu =====
```

1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT

```
-----  
Enter your choice (numeric) :1
```

```
===== Add =====
```

```
Enter a new account number: 9123456789
```

```
Please enter the name (first and last names): Yousef Ibrahim
```

```
Enter an Email address: Yousef@gmail.com
```

```
Enter a mobile number: 01014526385
```

```
Do you want to save your changes?  
(choose y or n)
```

In the case of choice number 2, that will send you to the **Delete** account part that will require you to enter the account number you want to delete consist of 10 digits, it will ask you if you want to save your changes in case of Yes just enter (y) and in case of you want to discard your changes enter (n).

```
===== Menu =====
1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT
-----
Enter your choice (numeric) :2

===== Delete =====

Enter an account number to be deleted: 9123456789

Do you want to save your changes?
(choose y or n)
```

In the case of choice number 3, that will send you to the **Modify** account part that will require you to enter the account number you want to modify consist of 10 digits, the new first and last name, new Email address and the new mobile number. At the end it will ask you if you want to save your changes in case of Yes just enter (y) and in case of you want to discard your changes enter (n).

```
===== Menu =====
1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT
-----
Enter your choice (numeric) :3

===== Modify =====

Enter an account number: 9123456789
Please enter a new name (first and last names): Ahmed Mohamed
Enter a new mobile number: 01200000005
Enter a new Email address: Ahmed@gmail.com

Do you want to save your changes?
(choose y or n)
```

In the case of choice number 4, that will send you to the **Search** part that will require you to enter the account number you want to search on consist of 10 digits and all the information of the account will be displayed on the screen.

```
===== Menu =====  
1. ADD  
2. DELETE  
3. MODIFY  
4. SEARCH  
5. ADVANCED SEARCH  
6. WITHDRAW  
7. DEPOSIT  
8. TRANSFER  
9. REPORT  
10. PRINT  
11. QUIT  
-----  
Enter your choice (numeric) :4  
  
===== Search =====  
  
Enter an account number: 9123456789  
Account Number: 9123456789  
Name: Ahmed Mohamed  
E-mail: Ahmed@gmail.com  
Balance: 0.00  
Mobile: 01200000005  
Opened : December 2023
```

In the case of choice number 5, that will send you to the **Advanced Search** part that will require you to enter a keyword like a part of name ex:(Robert) Note:(First litter should be in uppercase) and if any account name have a common litter will be displayed on the screen the screen if there is no account name with that keyword the program will say no result found.

```
Enter your choice (numeric) :5
```

```
===== Advanced Search =====
```

```
Enter a keyword: Robert  
Search results:
```

```
Account Number: 9700000001  
Name: John Roberto  
E-mail: j.roberto@outlook.com  
Balance: 100.00  
Mobile: 01009700001  
Opened : December 2008
```

```
Account Number: 9700000003  
Name: Michael Robert  
E-mail: michael@yahoo.com  
Balance: 300.00  
Mobile: 01009700003  
Opened : November 2008
```

```
Account Number: 9700000005  
Name: David Roberts  
E-mail: david123@gmail.com  
Balance: 400.50  
Mobile: 01009700005  
Opened : October 2015
```

In the case of choice number 6, that will send you to the **Withdraw** process it will require you to enter the account number you want to do the process on that consist of 10 digits, then it will ask you to enter the amount you want to withdraw but note that the maximum amount to withdraw is 10,000\$ and if the account balance can't afford that amount it will say insufficient quantity, At the end it will ask you if you want to save your changes in case of yes just enter (y) and in case you want to discard your changes enter (n).

```
===== Menu =====  
1. ADD  
2. DELETE  
3. MODIFY  
4. SEARCH  
5. ADVANCED SEARCH  
6. WITHDRAW  
7. DEPOSIT  
8. TRANSFER  
9. REPORT  
10. PRINT  
11. QUIT  
-----  
Enter your choice (numeric) :6  
  
===== Withdraw =====  
  
Enter the account number :9700000000  
Enter amount to withdraw: 500  
  
Do you want to save your changes?  
(choose y or n)
```

In the case of choice number 7, that will send you to **Deposit** process it will require you to enter the account number you want to do the deposit process on that consist of 10 digits, then it will ask you to enter the amount you want to deposit but note that the maximum amount to deposit is 10,000\$, At the end it will ask you if you want to save your changes in case of yes just enter (y) and in case you want to discard your changes enter (n).

```
===== Menu =====  
1. ADD  
2. DELETE  
3. MODIFY  
4. SEARCH  
5. ADVANCED SEARCH  
6. WITHDRAW  
7. DEPOSIT  
8. TRANSFER  
9. REPORT  
10. PRINT  
11. QUIT  
-----  
Enter your choice (numeric) :7  
  
===== Deposit =====  
  
Enter the account number :9123456789  
Enter amount to Deposit: 7999  
  
Do you want to save your changes?  
(choose y or n)  
y  
Changes saved successfully.  
-----  
  
Deposit process Successfully
```

In the case of choice number 8, that will send you to the **Transfer** process, in transfer process it will require you to enter the sender's account number that will send the money from his balance and to enter the receiver's account number that will get the money and if the sender balance can't afford the amount it will say insufficient quantity, At the end it will ask you if you want to save your changes in case of yes just enter (y) and in case of you want to discard your changes enter (n).

```

===== Menu =====

1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT
-----
Enter your choice (numeric) :8

===== Transfer =====

Enter sender's account number: 9123456789
Enter receiver's account number: 9700000000
Enter an amount to transfer: 5999

Do you want to save your changes?
(choose y or n)
y
Changes saved successfully.
Transfer successful!

```

In the case of choice number 9, that will send you to the **Report** part it will require you to enter the account number of the account you want to see the last 5 transaction happened if there is less than 5 it will be displayed too.

```

===== Menu =====

1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT
-----
Enter your choice (numeric) :9

===== Report =====

Enter Account Number: 9123456789
Deposit transaction +7999.00$
Transfer transaction sent 5999.00$ to 9700000000 Account Number

```

In the case of choice number 10, that will send you to the **Print** part, if you want to sort by name type (1), in case of sort by balance type (2), in case of sort by date type (3), you can sort the name from A to Z or from Z to A, you can sort the balance ascending or descending and you can sort the date from newer to older or older to newer.

Enter your choice (numeric) :10

===== Print =====

please enter the type of sort you want

- 1)sort by name
- 2)sort by balance
- 3)sort by date

1

please choose the order of sorting

- 1)A to Z
- 2)Z to A

1

Account Number: 9700000008

Name: Adam Mark

Email: ad.mark@gmail.com

Balance: 350.00

Mobile: 0100970008

Date Opened: 10-2015

Account Number: 9123456789

Name: Ahmed Mohamed

Email: Ahmed@gmail.com

Balance: 2000.00

Mobile: 01200000005

Date Opened: 12-2023

Account Number: 9700000006

Name: Daniel Graves

Email: dgrave@outlook.com

Balance: 450.00

Mobile: 01009700006

Date Opened: 1-2020

Enter your choice (numeric) :10

===== Print =====

please enter the type of sort you want

- 1)sort by name
- 2)sort by balance
- 3)sort by date

1

please choose the order of sorting

- 1)A to Z
- 2)Z to A

2

Account Number: 9700000002

Name: Timothy Korman

Email: t.korman@gmail.com

Balance: 200.00

Mobile: 01009700002

Date Opened: 12-2015

Account Number: 9700000004

Name: Roberto Thomas

Email: rob.thomas@gmail.com

Balance: 400.50

Mobile: 01009700004

Date Opened: 11-2015

Account Number: 9700000007

Name: Philipe Brian

Email: p.brian@outlook.com

Balance: 460.00

Mobile: 01009700007

Date Opened: 2-2020

Enter your choice (numeric) :10

===== Print =====

please enter the type of sort you want

- 1)sort by name
- 2)sort by balance
- 3)sort by date

2

please choose the order of sorting

- 1)Accending
- 2)Decending

1

Account Number: 9700000001

Name: John Roberto

Email: j.roberto@outlook.com

Balance: 100.00

Mobile: 01009700001

Date Opened: 12-2008

Account Number: 9700000002

Name: Timothy Korman

Email: t.korman@gmail.com

Balance: 200.00

Mobile: 01009700002

Date Opened: 12-2015

Enter your choice (numeric) :10

===== Print =====

please enter the type of sort you want

- 1)sort by name
- 2)sort by balance
- 3)sort by date

2

please choose the order of sorting

- 1)Accending
- 2)Decending

2

Account Number: 9700000000

Name: Michael Jones

Email: m.jones@gmail.com

Balance: 6999.00

Mobile: 01009700000

Date Opened: 12-2007

Account Number: 9123456789

Name: Ahmed Mohamed

Email: Ahmed@gmail.com

Balance: 2000.00

Mobile: 01200000005

Date Opened: 12-2023

Account Number: 9700000007

Name: Philipe Brian

Email: p.brian@outlook.com

Balance: 460.00

Mobile: 01009700007

Date Opened: 2-2020

<pre> ----- Enter your choice (numeric) :10 ===== Print ===== please enter the type of sort you want 1)sort by name 2)sort by balance 3)sort by date 3 please choose the order of sorting 1)Oldest first 2)Newest first 2 Account Number: 9123456789 Name: Ahmed Mohamed Email: Ahmed@gmail.com Balance: 2000.00 Mobile: 01200000005 Date Opened: 12-2023 Account Number: 9700000007 Name: Philipe Brian Email: p.brian@outlook.com Balance: 460.00 Mobile: 01009700007 Date Opened: 2-2020 Account Number: 9700000006 Name: Daniel Graves Email: dgrave@outlook.com Balance: 450.00 Mobile: 01009700006 Date Opened: 1-2020 </pre>	<pre> ----- Enter your choice (numeric) :10 ===== Print ===== please enter the type of sort you want 1)sort by name 2)sort by balance 3)sort by date 3 please choose the order of sorting 1)Oldest first 2)Newest first 1 Account Number: 9700000000 Name: Michael Jones Email: m.jones@gmail.com Balance: 6999.00 Mobile: 01009700000 Date Opened: 12-2007 Account Number: 9700000003 Name: Michael Robert Email: michael@yahoo.com Balance: 300.00 Mobile: 01009700003 Date Opened: 11-2008 Account Number: 9700000001 Name: John Roberto Email: j.roberto@outlook.com Balance: 100.00 Mobile: 01009700001 Date Opened: 12-2008 </pre>
---	--

In the case of choice 11, the program will close.

```

===== Menu =====

1. ADD
2. DELETE
3. MODIFY
4. SEARCH
5. ADVANCED SEARCH
6. WITHDRAW
7. DEPOSIT
8. TRANSFER
9. REPORT
10. PRINT
11. QUIT
-----
Enter your choice (numeric) :11
Quitting the program ....

```

Features explanations and Algorithms

- **Login:**

1. Opens the users file and if not found it presents an error.
2. Reads the number of lines in the file.
3. Uses that number to memory allocate an array of structs for users.
4. It asks the user to input username and password and checks if it is found or not and if not found it loops again to ask for input again till successful login.

- **Load Account:**

1. Initialization:
Sets the variable numAccounts to zero to prepare for loading new accounts.
2. File Opening:
Opens the "accounts.txt" file in read mode.
If the file doesn't exist or there's an issue opening it, the function displays a "File not found" message and exits the program.
3. Reading File Contents:
Utilizes a loop to read data from the file until the end of the file (EOF) is reached.
Reads formatted data from each line of the file using fscanf.
Parses and assigns the read values to the respective fields in the account_data array, such as account number, name, email, balance, mobile number, and date opened.
Increments the numAccounts variable to keep track of the number of successfully loaded accounts.
4. File Closure:
Closes the file to release resources and ensure data integrity.

- **Add account:**

1. It asks the user to input account number and checks if it is valid and not already existing.
2. It asks the user to input the name which expects 2 names and then checks if the name is valid or not and if valid it sends the name to a function to make the name in the same format as the other names saved in the accounts file.
3. It asks the user to input email and then checks if the email is valid or not and if not loops till valid email is entered.
4. It asks the user to input mobile number and then checks if it's valid or not and if not loops till valid mobile number is entered.
5. Sets the balance of the new account to zero.
6. Gets the current time using a function.

7. All these are saved in the array of struct of accounts after the last account and increments the number of accounts.
8. Asks user if he wants to save his actions to write it in the file and if he doesn't save the file is loaded is again to set the number of accounts to before the add function.

- **Delete account:**

1. It asks the user to input account number and checks if it is valid.
2. Checks if the account exists or not and if it exists saves its index in the array in a variable.
3. Checks if the account balance isn't zero to delete.
4. Swaps the account data of the last account in the array with the data of the found account and then decrements the number of accounts.
5. Asks user if he wants to save his actions to write it in the file and if he doesn't save the file is loaded is again to set the number of accounts to before the delete function and to fill the array again with order before the delete function.

- **Modify Account:**

1. It asks the user to input account number and checks if it is valid.
2. Checks if the account exists or not and if it exists saves its index in the array in a variable.
3. If the account is found it enters a loop to modify the account data (name, email, mobile number).
4. Save changes.

- **Save feature:**

1. Asks the user to choose between 'y' or 'n' whether to save or discard the changes made by the user.
2. Reads user input into 'ans' If 'ans' is in uppercase, converts it to lowercase.
3. there are two cases for the choice by the user
first case (if the user chooses y)
Opens the file called accounts.txt in write mode.
If the file is not found displays "Save failed due to file not found."
Return 0 value which means change is unsuccessful.
If the file is found
For each account_data entry, write account details to the file then closes the file and displays "Changes saved successfully." then load accounts from file
Return 1 which means change successful.

4. second case (if the user chooses n) load accounts from file to disregard any changes made by user display "Changes discarded." return 0 which means no changes happened.
5. otherwise display "Invalid input." and make a recursive call to save function to handle invalid input
return 0 for default return in case of unexpected termination.

- **Query (Search):**

- 1-first the user will enter infinite loop where he will enter the account number he want to search for it will be stored in string acc_num
- 2-acc_num will be sent to function named is_valid_acc_num that make sure that the user entered a valid account number if not valid it will let him enter again because of the infinite loop till entering it correctly
- 3-if it is valid it will enter the account number correctly it will enter a for loop where i is the variable that will be used to call every account number till acc_num (the of account stored)
- 4-in the for loop there is an if condition that used strcmp to compare the acc_num(the in put string of the user) with the ever account number the variable i will call if true it will the account details of the account and break it there no matching it will exist the loop
- 5-it will enter if condition if i= the number of account this mean that the previous if condtion didnt find any matching account number

- **Advanced search:**

1. Initialization:
Declare variables keyword (to store the user-provided search keyword), count (to track matching characters), and z (for iterating through accounts).
Print a header for the advanced search.
2. User Input:
Use a do-while loop to prompt the user to enter a keyword.
Validate the keyword using the is_invalid_name function (which is used to check for invalid inputs).
3. Search Results:
Use a while loop to iterate through accounts (z represents the current account index).
Use nested loops to compare each character of the keyword with the corresponding characters in the account name.
If a match is found, increment the count variable.

If the count matches the length of the keyword, it means a complete match is found.

Print the details of the matching account.

Reset counters and move to the next account.

If the end of the account name is reached, reset counters and move to the next account.

4. Display Results:

If no matches are found (flag is not set), display a message indicating that no matches were found.

- **Withdraw:**

1. Ask the user to enter the account number.
2. it searches the entered account number in array using the function (findAccountIndex).
3. asks the user to enter the amount of withdraw.
4. checks if the withdrawal amount exceeds the maximum limit (which is 10000\$), is less than or equal zero or if it is greater than the account balance.
5. If the conditions are true, update the account balance, save the transaction to a file and then prints if success or failure.~

6. **In the find the account index:~**

Searches for the index of a given account number in an array of the accounts ('account_data').

Returns the index of the account number is found or -1 if not found.

- **Deposit:**

1. Ask the user to enter the account number
2. it searches the entered account number in array using the function (findAccountIndex).
3. asks the user to enter the amount of deposit.
4. checks if the withdrawal amount exceeds the maximum limit (which is 10000\$), is less than or equal zero.
5. If the conditions are true, update the account balance, save the transaction to a file and then prints if success or failure.

- **Transfer:**

1. Input Handling:

Asks for the sender's and receiver's account numbers.

Validates the account numbers (must be 10 digits).

Checks if both sender and receiver accounts exist in the system.

2. Amount Confirmation:

Requests the user to input the amount to transfer.

Verifies the validity of the entered amount:

Checks if the amount is numeric and positive.

Validates if the sender's balance covers the transfer.

3. Transfer Process:

Deducts the transferred amount from the sender's balance.

Adds the transferred amount to the receiver's balance.

Records the transaction details in the respective account files for tracking.

4. Error Handling:

If there's any issue with recording the transaction, it displays an error message and stops.

If the transfer is successful, it shows a success message.

5. File Operations:

Opens the sender and receiver account files to append the transaction details.

If there's an error opening the files, it displays an error message and stops.

• **Report:**

1. User Input:

Asks the user to input an account number.

Checks if the entered account number is valid (must be 10 digits). If not, prompts the user again.

2. Account Check:

Searches for the entered account number.

If the account number is not found, it displays a message and stops.

3. File Opening:

Constructs the file name by adding ".txt" to the account number.

Tries to open a file with that name in read mode.

If the file doesn't exist or can't be opened, it shows a message and ends.

4. Reading Transactions:

Reads the last 5 lines of the file (or less if there are fewer transactions) and stores them.

5. Display Transactions:

Shows the stored transactions in the console:

If there are fewer than 5 transactions, it displays all of them.

If there are more than 5 transactions, it shows the last 5 in reverse order.
then closes the file after reading.

- **Print sort:**

1. This function provides an interface for the user to choose the type of sorting they want (by name, balance, or date) and then calls the corresponding sorting function (sortByName(), sortByBalance(), or sortByDate()). It ensures that the user input is valid and repeatedly prompts until a valid choice is made.

2. sortByDate()

This function sorts an array of accounts based on the date they were opened, either in ascending or descending order. It uses the bubble sort algorithm to compare and swap accounts based on their opening dates. After sorting, the user is prompted to choose the order (oldest first or newest first) and then display the sorted results.

3. sortByBalance()

Similar to sortByDate(), this function sorts the array of accounts based on their balances in either ascending or descending order. It uses the bubble sort algorithm to compare and swap accounts based on their balances. After sorting, the user is prompted to choose the order (ascending or descending) and then display the sorted results.

4. sortByName()

This function sorts the array of accounts alphabetically by name, either in ascending (A to Z) or descending (Z to A) order. It uses the bubble sort algorithm to compare and swap accounts based on their names. After sorting, the user is prompted to choose the order (A to Z or Z to A) and then display the sorted results.

5. These functions collectively offer a sorting and printing mechanism for the array of accounts, providing flexibility in sorting criteria and order. The sorting algorithm used is a basic bubble sort, and the user is given options to control the sorting order.